

Tabular Foundation Models

A new way to get predictions from the rows and columns
that run your business.



Every team has questions hiding in its data

?

Which customers are likely to churn?

?

What is a customer worth over time?

?

Which transactions look like fraud?

?

How much demand is coming next quarter?

?

Which leads deserve attention first?

?

Which products will perform best?



The problem is rarely a lack of data. It's turning data into decisions.

Most organizations are rich in structured data. The hard part has always been getting from a full spreadsheet to a confident answer.



Four parts

01 The foundation-model shift

02 The data that runs your business

03 What tabular foundation models are

04 From models to answers



A separate model for every task

THE WAY IT USED TO WORK

Build one model per problem

Detect fraud?

Train a fraud model.

Forecast sales?

Train a forecast model.

Sort feedback?

Train a text model.



THE FOUNDATION-MODEL IDEA

Learn general patterns first

Train one model on a huge amount of data so it learns patterns it can reuse. Then apply it to many different tasks, instead of building from zero every time.



Learn the patterns once. Apply them everywhere.



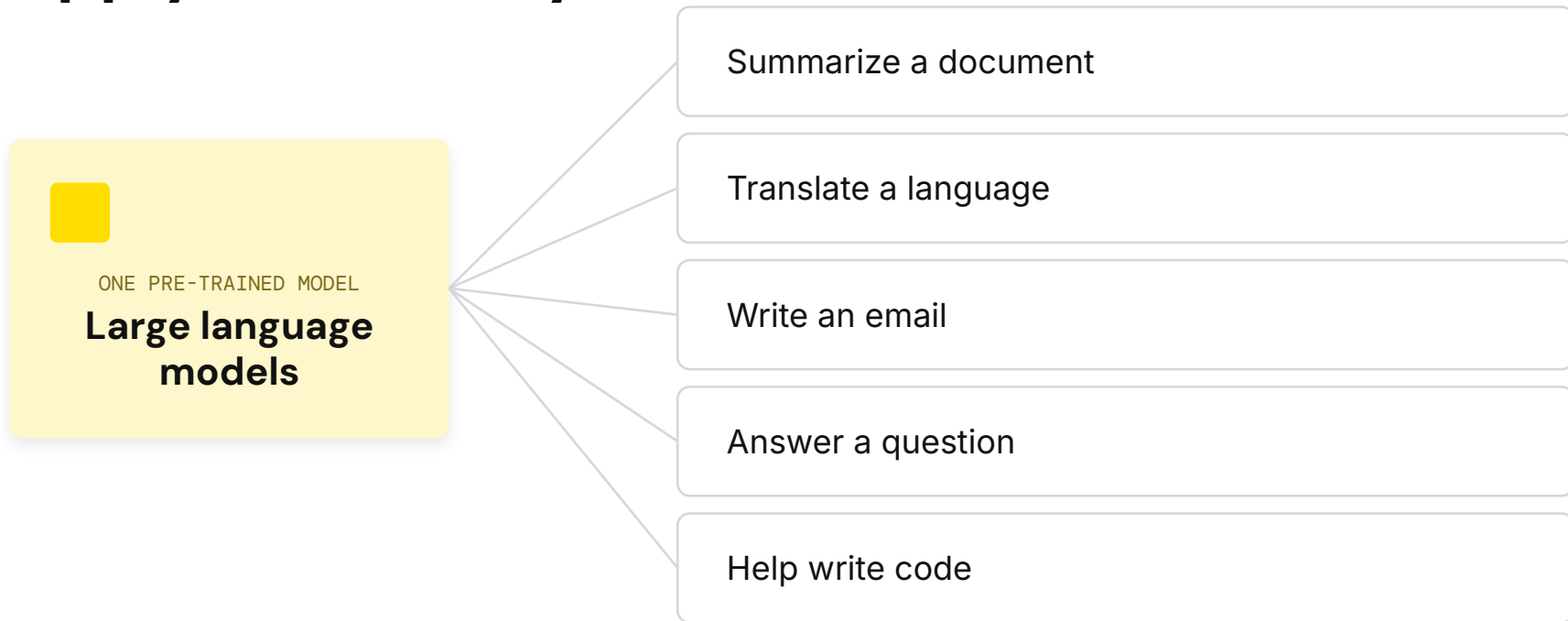
ONE PRE-TRAINED MODEL

**Large language
models**

One model, trained once on a huge amount of text, before it is ever pointed at a specific job.



Learn the patterns once. Apply them everywhere.



You describe what you want.
The model already knows how.

No training run for each new job. It draws on patterns it learned once, up front.



AI got easy for words. Business data stayed hard.

ASK IT TO WRITE A REPORT

Done in

seconds

PREDICT FROM A SPREADSHEET

Still takes

weeks

Language tasks feel instant. Predicting from business data often still follows a workflow designed years ago.



You've worked with this before

CUSTOMER	AGE	COUNTRY	PLAN	ANNUAL REVENUE	YEARS	CHURNED?
C-1042	34	USA	Pro	\$2,400	3	No
C-1043	51	UK	Basic	\$540	1	Yes
C-1044	29	Canada	Pro	\$3,180	5	No

Every row is a record. Every column is something you already know.



You've worked with this before

CUSTOMER	AGE	COUNTRY	PLAN	ANNUAL REVENUE	YEARS	CHURNED?
C-1042	34	USA	Pro	\$2,400	3	No
C-1043	51	UK	Basic	\$540	1	Yes
C-1044	29	Canada	Pro	\$3,180	5	No
C-1045	43	Germany	Team	\$8,900	4	?
C-1046	38	Spain	Basic	\$620	2	?



Known & unknown today



A prediction (coming next lesson)

One column is the one you want to **predict**.

Every row is a record. Every other column is something you already know about it. The model learns how those known columns relate to the one you care about.



Spreadsheets are the language of business



Banks

CREDIT RISK



Retail

DEMAND PLANNING



Insurance

CLAIMS



Healthcare

OUTCOMES



Manufacturing

QUALITY



Airlines

PRICING



Marketing

TARGETING



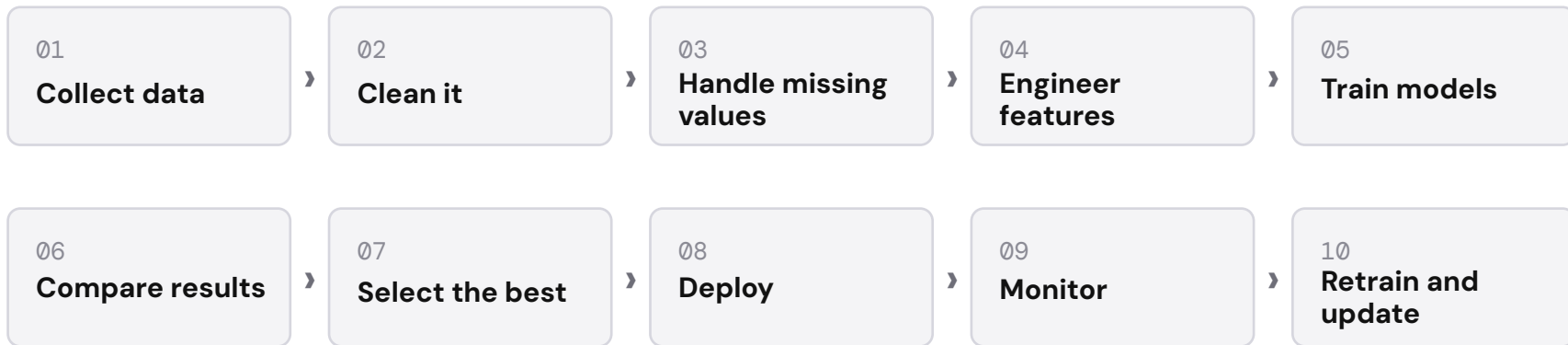
Finance

FORECASTING

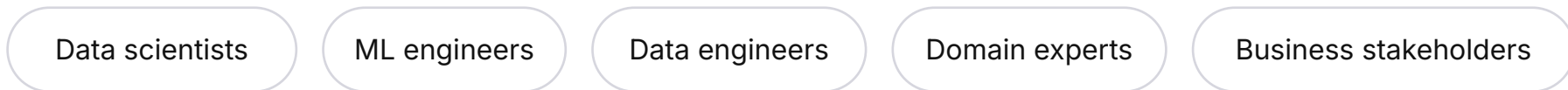
Numbers, dates, categories, locations, behaviour. Most of the world's business data lives in rows and columns.



From a question to a prediction, the long way



AND THE PEOPLE IT USUALLY TAKES



Many business questions **never make it through the process.**

Sometimes the problem is too small to justify a full project.

Sometimes the resources aren't there.

Sometimes people just need an answer faster than the workflow allows.



Language models learn text. These learn tables.

LANGUAGE MODEL

Learns from text

Relationships between words. Grammar. Context. The meaning that connects a sentence together.

TABULAR FOUNDATION MODEL

Learns from tables

Relationships between columns. How values interact. The patterns that show up again and again across structured datasets.



It doesn't start from zero

TRADITIONAL ML

Starts from scratch

Every new problem begins with a blank slate. Gather data, build features, train a model, and start over for the next question.



 TABULAR FOUNDATION MODEL

Starts with prior knowledge

It already understands common structures and relationships in business data. So it adapts to a new prediction task far faster.



Who spots the pattern faster?

FIRST DAY ON THE JOB

**Has never read a business
report**



TWENTY YEARS IN

**Has read thousands, across
many industries**



The experienced analyst wins, not because they've seen this exact spreadsheet, but because they've seen thousands like it. **Tabular foundation models work the same way.**



**It has already seen thousands of
tables like yours.**

The expert doesn't win by seeing this exact spreadsheet. A tabular foundation model brings that same accumulated experience to a dataset it has never met.



Everyday jobs on business data



Classification

Will it happen?

Churn, loan default, fraud. Predict a yes-or-no outcome.



Regression

How much?

Sales, price, lifetime value. Predict a number.



Missing values

Fill the gaps

Infer likely values from the patterns already in the data, with less manual guesswork.



Hidden patterns

Surface the links

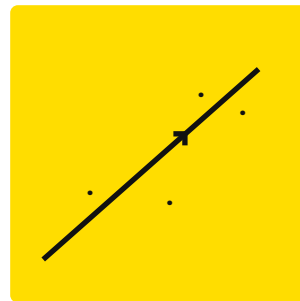
Spot relationships across columns that are hard to see by hand.



Regression

How much? Predict a number.

Tabular foundation models can also estimate numerical values, learning how the known columns point to the figure you need.



COMMON REGRESSION QUESTIONS

Forecast future sales

Predict product prices

Estimate lifetime value

Score risk

Project revenue



Real data is rarely perfect

If you've ever worked with spreadsheets, you know the gaps are everywhere.



Gaps in columns

A field left blank



Incomplete values

Only part of the picture



Missing records

Whole rows absent

Traditionally, filling these in meant manual work and educated guesses.



Fill the gaps from **patterns** **already in the data.**

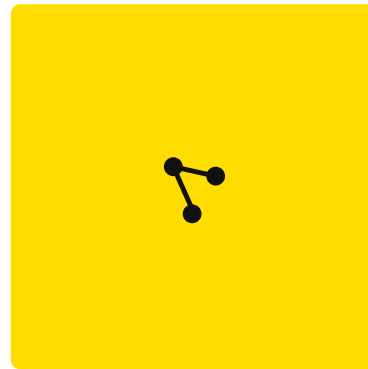
A tabular foundation model can infer the likely value from everything else in the dataset, so teams move faster with less guesswork.



Hidden patterns

Links across columns you would miss by hand.

These models can surface relationships across columns that are hard to spot through manual analysis, connections between fields you might never think to compare.



Not just a single prediction. A read on how the pieces of your data actually relate.



A SIMPLER WAY TO WORK WITH DATA

The biggest change isn't what they predict. It's how you work with predictive AI.



Building a model from scratch

Generating a prediction the old way meant working through all of this first:

Prepare data

Create features

Select algorithms

Tune hyperparameters

Evaluate results

Manage deployment

Valuable work. Also a lot to get through before anyone sees a single answer.



Start from **prior knowledge**, not a blank slate.

A tabular foundation model reuses what it learned during pre-training and applies it to your new task. The result is a faster path from data to insight.



Less plumbing. More thinking.

LESS TIME SPENT ON

- Building models
- Engineering features by hand
- Moving data between tools

MORE TIME SPENT ON

**Understanding the
business problem
itself**



STILL TRUE

This doesn't replace machine learning.

Expertise still matters, and traditional workflows are still the best choice for many problems. Tabular foundation models widen the door. They don't close the old one.



More people can ask predictive questions

Not only machine-learning specialists. Anyone close to the business can start exploring predictive questions.

Analysts

Business users

Operations

Finance

Marketers

Decision-makers

The expertise doesn't disappear. The starting line just moves closer to the question.



**The focus moves from
building models to solving
problems.**



Foundation models reshaped language and images. Now they're reaching your data.

Tabular foundation models apply the same principles to the rows and columns that run most organizations, learning from large collections of tabular datasets.



More accessible predictions.

Less complexity. A faster path to insight.

The technology is still evolving, and it's already drawing serious attention across the AI industry.



So how can a model skip all those steps?

?

If traditional ML needs data prep, features, training, evaluation, and deployment, how does a foundation model simplify so many of those steps?

?

And if it isn't trained from scratch every time, how does it generate predictions for a dataset it has never seen?

That's exactly what the next lesson is about.



The focus shifts to the problem

FROM

Building models



TO

Solving the business problem

WHEN TRADITIONAL ML STILL WINS

Traditional machine learning still wins when it counts: heavy regulation, massive scale, the last percent of accuracy, and deeply custom features. Tabular foundation models widen the door. They don't close the old one.



Next: how they actually work

A closer look at how a tabular foundation model generates predictions for a dataset it has never seen before.

Words got easy

→ Tables stayed hard

→ Models generalize

→ Tables are next

TABULAR FOUNDATION MODELS

Making Sense of Your Data Faster

From what these models are, to what they change in everyday work.



Last lesson, in three ideas

1

A shift in AI

Foundation models
reshaped language, images
and content.

2

Now for tables

The same idea is reaching
structured business data.

3

Where decisions live

Tabular data still drives
most business calls.

?

**But what does this actually mean
in practice?**

Three questions we'll answer

1

Where

Where can tabular foundation models help?

2

Why

Why are teams investing in them?

3

What changes

What shifts in day-to-day work?

What's ahead

01 Where prediction earns its keep

02 From models to questions

03 When answers get easy

04 Closing the prediction gap



**Leaders rarely ask for models.
They ask for
answers.**

What leaders actually ask

? Which customers will leave?

? Which deals will close?

? What should we stock next?

? Which transactions to check?

**These are all
prediction problems.**

Customer churn

Who is most likely to cancel next?

Usage, tenure and support history hold the signal.
Spotting risk early means acting before revenue is lost.

A churn prediction, in familiar rows

Customer	Tenure	Support tickets	Churn risk
Nord Retail	14 mo	2	Low
Vela Foods	3 mo	9	High
Arqa Bank	8 mo	?	Medium

 predicted inferred gap

Forecasting

How much should we plan for?

Inventory, demand, headcount and revenue. The better the read on what's ahead, the smarter the planning.

Risk and fraud

Where should attention go first?

Not every transaction needs the same scrutiny. Risk scores point effort to where it matters most.

Filling the gaps

What belongs in the blanks?

Real data is rarely complete.

The model infers likely values from patterns already in the data.

Different problems, one move



Customer churn



Forecasting



Risk & fraud



Data completion

**Use what you already have
to see what's
likely next.**

?

**These use cases aren't new.
So what's
different?**

One problem, one model

Predict churn → build a churn model

Forecast revenue → build a forecasting model

Score risk → build a risk model

**More models. More pipelines.
More to
maintain.**

The question changes

BEFORE

**Which model
should we build?**



NOW

**What would we
like to predict?**

**The work moves from building
models to generating
insight.**

Data scientists still choose the approach



This adds a tool to the toolkit. It doesn't remove the expertise that decides which approach fits a problem.

How predictive projects used to flow

**Business
finds a
problem**



**Handed to a
technical
team**



**Data
collected &
prepared**



**Models built
& evaluated**



**Results
returned**

**It works. It also creates
delays and bottlenecks.**

Two worlds, a lot of coordination

BUSINESS

**Understands
the problem**



TECHNICAL

**Understands
the models**

the gap in between

More people can take part

1

Analysts

Explore questions more independently.

2

Business users

Experiment without a full project.

3

Technical teams

Spend time on the harder problems.

**From “what algorithm?”
to
“what **decision**?”**

**Curiosity is limited by effort.
People stop asking when
answers get hard.**

Questions rarely explored

1

Sales

Do engagement metrics predict renewals?

2

Finance

Is there a better forecasting approach?

3

Operations

Any early signs of supply disruption?

Every question became a project

More meetings

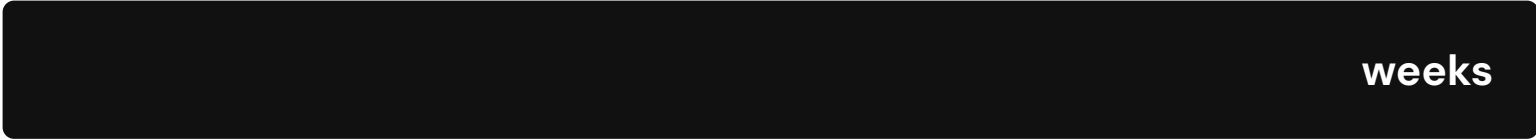
More data preparation

More development cycles

More waiting

From weeks of setup to a quick test

Traditional ML project



weeks

Foundation-model test



minutes

**When asking gets easier,
teams ask more.**

Predictive AI stops being specialist-only.

The people closest to the problem



Marketing



Sales



Operations



Finance



**Customer
success**



Product

The goal isn't to replace expertise



You still need to know which data carries real predictive signal.
Exploring those ideas just gets much faster.

**For H2O.ai, this was a
natural area to explore.**

Built on years of structured-data work

1

The people

Kaggle Grandmasters and
AI researchers.

2

The problems

Churn, forecasting, fraud
and risk.

3

The focus

Value from tabular
business data.

From custom models to foundation models

Custom models

›

AutoML

›

Foundation models

**One model learned broad
patterns and applied them
anywhere.**

?

**Could the same idea
simplify prediction?**

The direction is clear

1

Fit in

Work inside the tools
teams already use.

2

Move faster

Go from data to decisions
sooner.

3

Reach more

Open access beyond
specialist teams.

**A prediction has little value
until someone acts on it.**

The value is in the next step

 **A CSM reaches out early**

 **Finance adjusts the forecast**

 **An analyst checks a flag**

 **Ops prepares for disruption**

Most tools explain yesterday

DESCRIPTIVE

**What
happened?**



PREDICTIVE

**What's
likely next?**

**questions
we'd ask**

THE GAP

**questions
we can answer**

**The gap between the questions we'd ask
and the ones we can afford to answer**

**Foundation models can
narrow that gap.**

People work where the data already is

Spreadsheets, CRMs, dashboards, ERPs.

The tools that win aren't the most advanced. They're the ones that fit into work people already do.

**The best AI fits into work
people **already do.****

**This isn't about technology.
It's about **access**.**

**From understanding
yesterday to preparing for
tomorrow.**

NEXT LESSON

From the why to the how

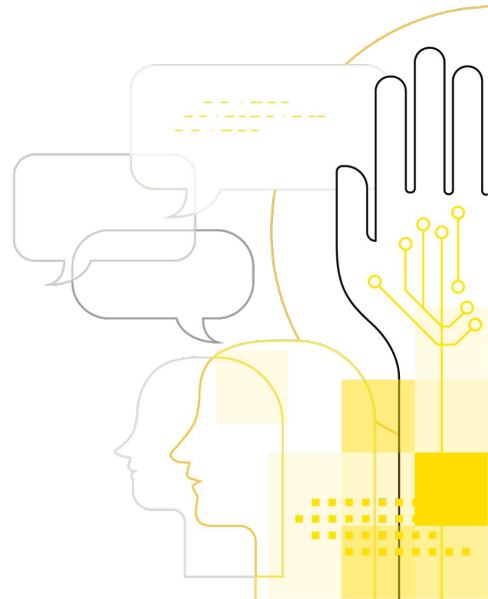
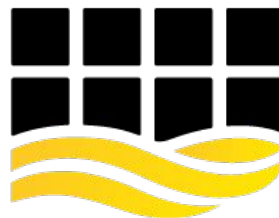
Next, we put this to work in a spreadsheet with **TabH2O**.
Familiar data in, predictions out, in a few steps.

Where it helps

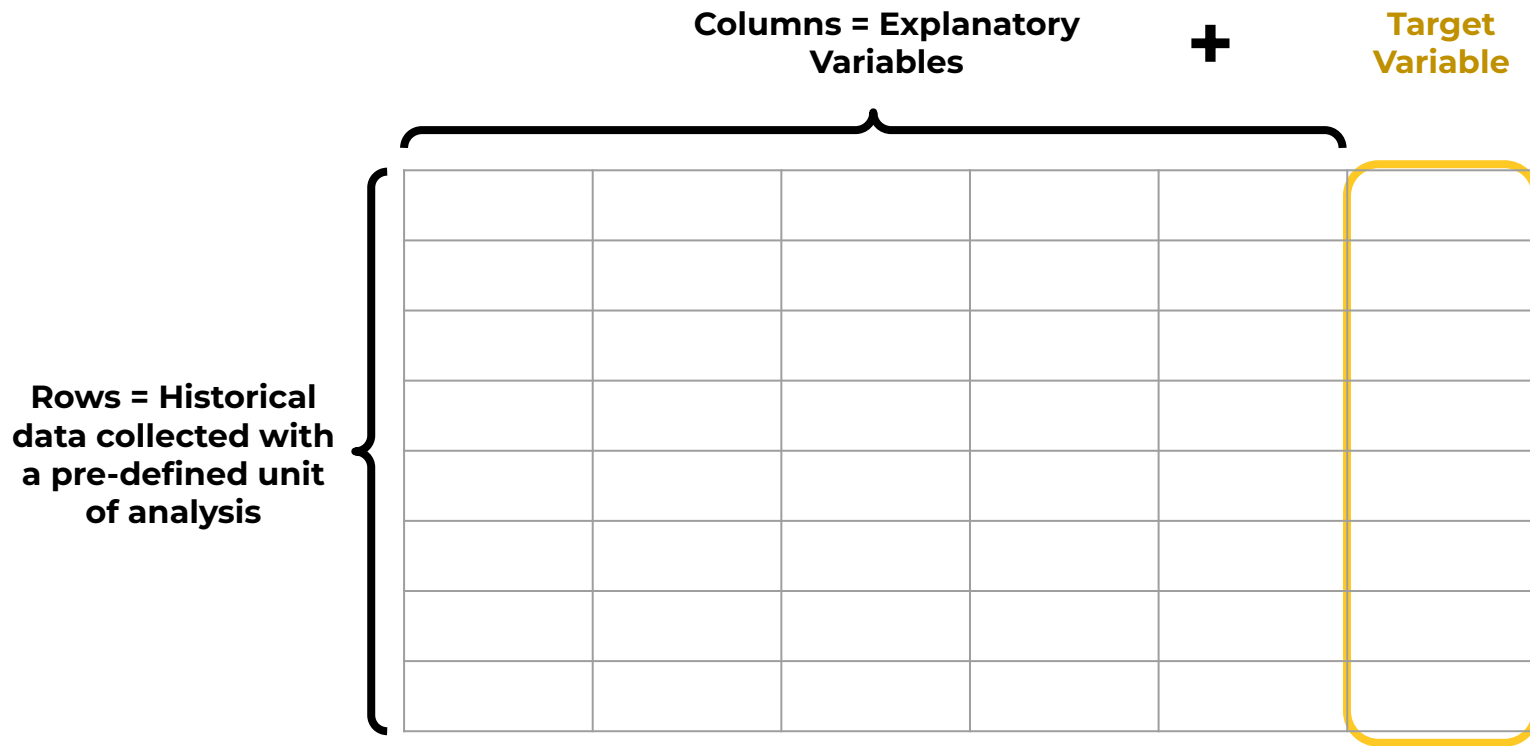
What changes

Why access wins

Predict Your Data in Excel with **TabH2O**

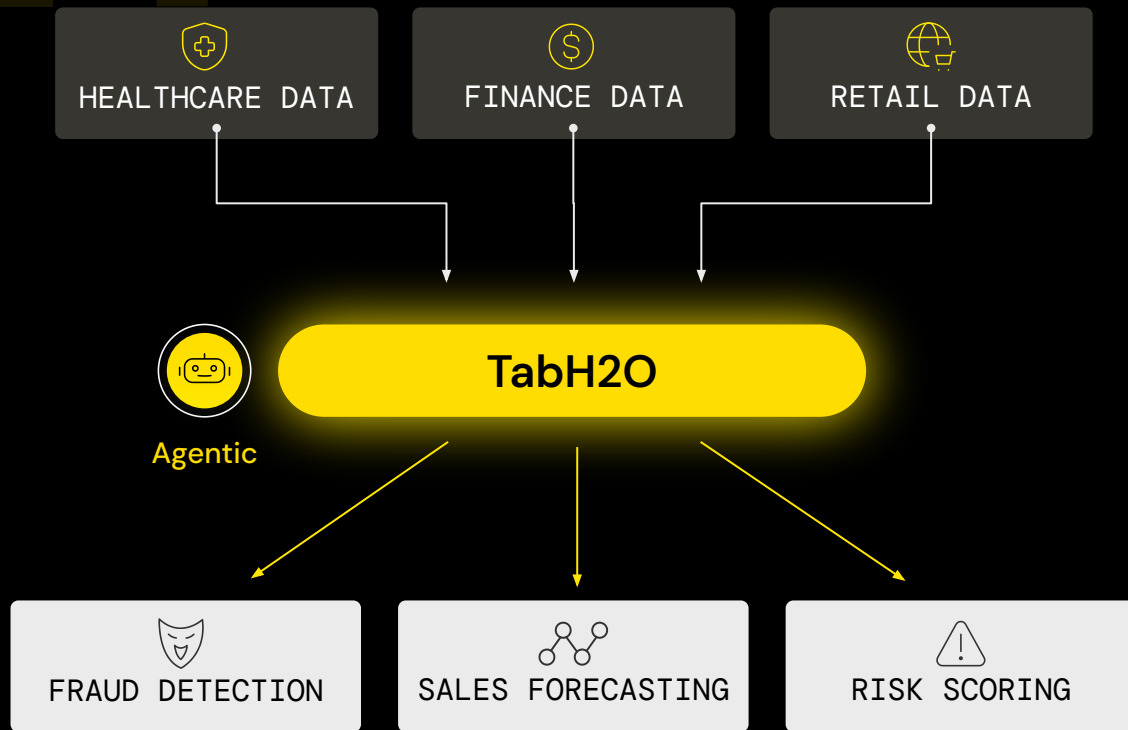


The Tabular Structured Data Format



TabH2O > H2O.ai proprietary tabular foundation model

H2O.ai



- **Single model for all use cases**

No training, no fine-tuning, no feature engineering

- **Works across domains**

Finance, healthcare, e-commerce and many more

- **No hidden biases**

Pre-trained on millions of synthetic datasets - doesn't use semantics

- **State of the art technology**

Outperforming best in class ML models

Common Data Issues in Business Datasets



Missing & Incomplete Records



Imbalanced Data



High-Cardinality Categories



Mixed Data Types & Noisy Data



Outliers & Prediction Gaps

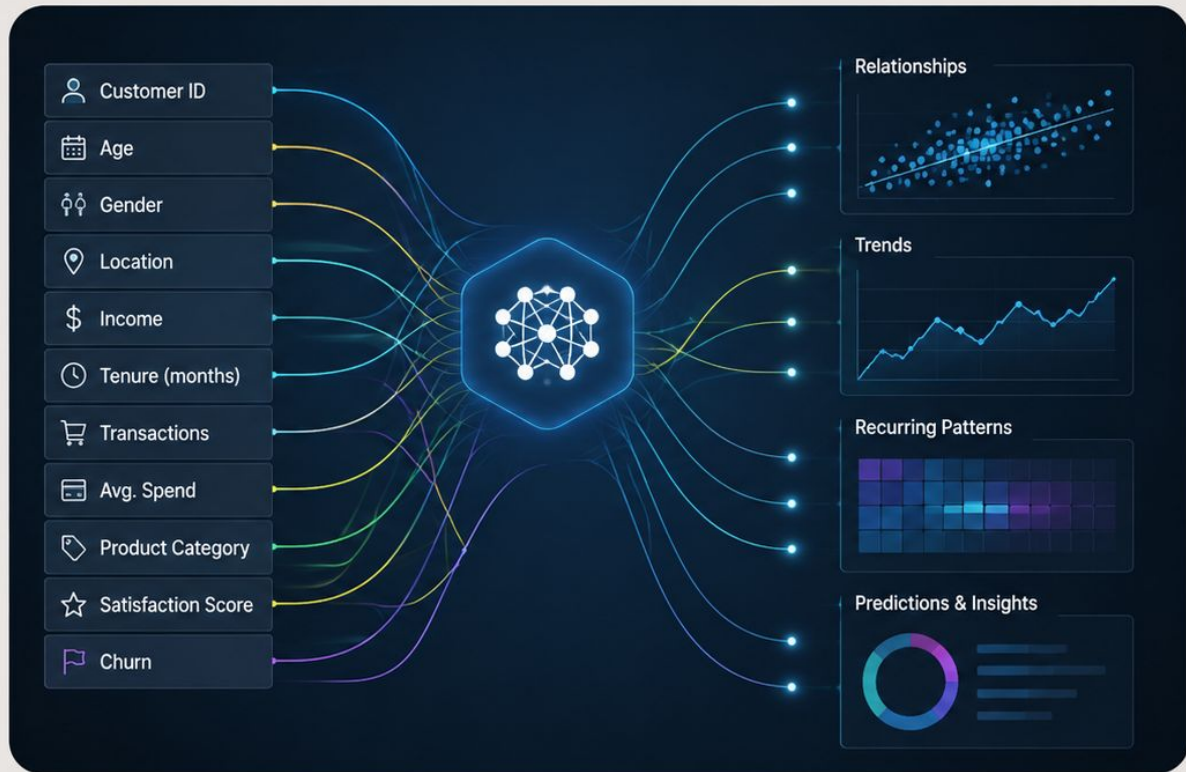
The TabH2O Advantage

Uncover patterns in imperfect business data to generate contextual predictions with **minimal preprocessing**.

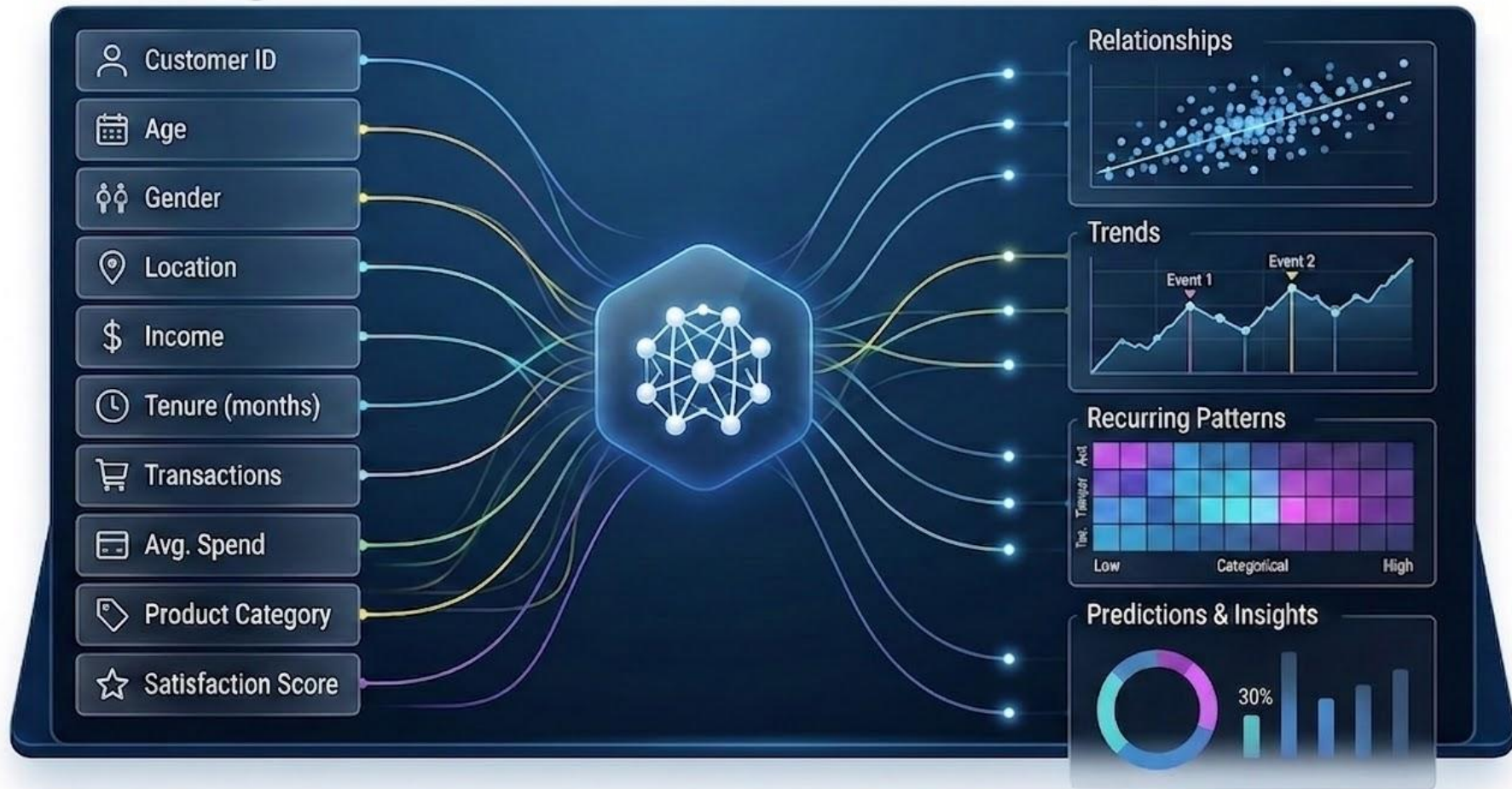
Learning Patterns from Tabular Data

TabH2O is built on a pre-trained **tabular foundation model** that has already learned common patterns found across structured datasets.

This includes **relationships** between columns, recurring **trends**, and typical behaviors often present in spreadsheets, databases, and business tables.



Learning Patterns from Tabular Data



Making Predictions: Labeled vs. Unlabeled Data



Every time you want to generate predictions, you provide a set of **labeled rows**, meaning rows where the outcome is already known, and a set of **unlabeled rows**, where the outcome is missing.

Labeled Data

(Outcome is known)

Transaction	Customer	Age	Gender	Amount	Is Fraud?
1	C001	34	F	\$120.00	No
2	C002	45	M	\$300.00	No
3	C003	29	F	\$89.00	No
4	C004	37	M	\$450.00	Yes
5	C005	52	F	\$200.00	No
...	...	...	...	...	...



Used to **train** the model
(Model learns patterns)

Unlabeled Data

(Outcome is missing)

Transaction	Customer	Age	Gender	Amount	Is Fraud?
101	C101	41	M	\$220.00	?
102	C102	28	F	\$150.00	?
103	C103	33	M	\$310.00	?
104	C104	47	F	\$95.00	?
105	C105	36	M	\$600.00	?
...	...	...	...	...	...



Machine Learning
Model



Predictions



Used to **generate predictions**
(Model applies learned patterns)



The model learns from the labeled data and applies those patterns to the unlabeled data to predict the missing outcomes.

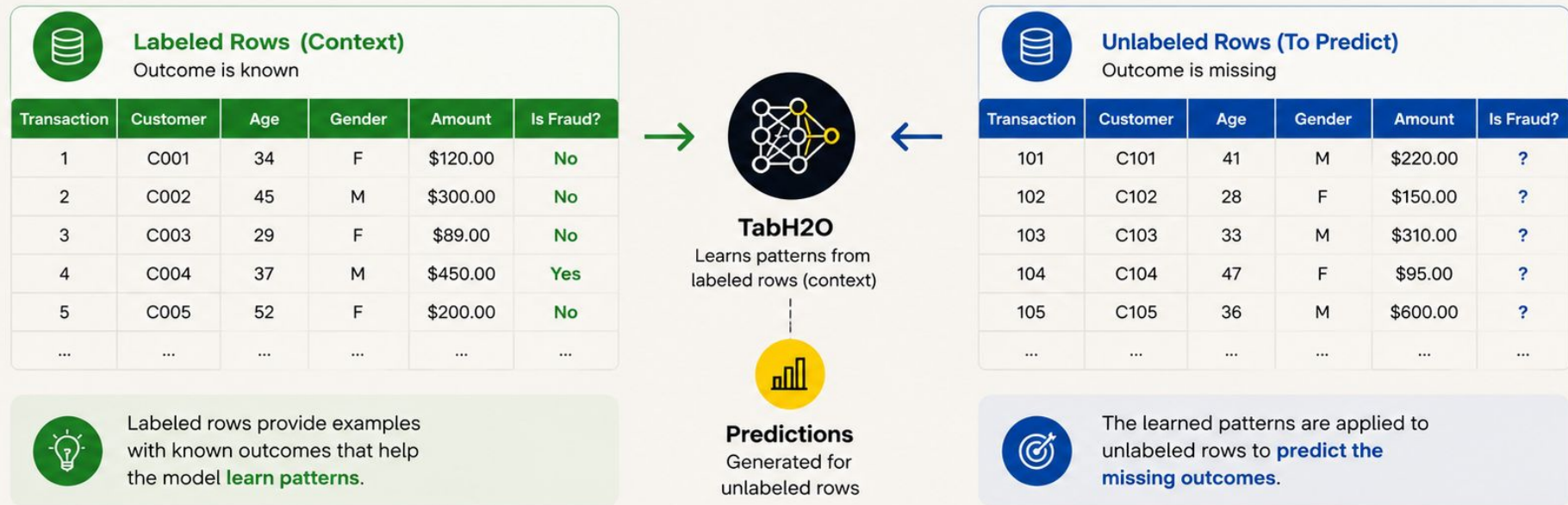
Making Predictions:

Labeled Rows vs. Unlabeled Rows

H2O.ai



TabH2O uses the **labeled rows as context** each time you run it to understand the patterns in your data, and **generates predictions** for the unlabeled rows accordingly.



The better and more representative your **labeled rows**, the more accurate and reliable your predictions will be.

Use Your Own Data or Follow Along with Our Sample Dataset



Use your own data

Bring your own tabular business data to test TabH2O on real-world prediction tasks.



Follow along step by step

Or use our sample customer churn dataset to reproduce the workflow demonstrated in this video.



Toy dataset available

For this demo, we've also made a small toy dataset available in our GitHub repository.

sample_customer_churn (Preview)

ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	...	PAY_AMT1	PAY_AMT2	PAY_AMT3	PAY_AMT4	PAY_AMT5	PAY_AMT6	default payment next month (Target)
24001	50000	1	2	2	23	...	8	2401	2254	2004	704	707	0
24002	60000	1	2	2	26	...	2282	2324	2049	2000	3000	1120	1
24003	400000	1	2	2	27	...	2501	10009	1437	1105	510	959	0
24004	20000	1	5	2	27	...	0	0	0	900	0	0	0
24005	50000	1	3	2	27	...	0	100	0	70	200	100	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
24021	360000	1	1	2	24	...	1003	1803	777	-3	-3	-3	0
24022	60000	1	1	2	25	...	5914	61210	52757	29209	26403	26969	1
24023	80000	1	2	2	25	...	0	78091	80288	82150	319	85	0



Download the latest dataset here:

github.com/h2oai/h2o-university-resources/blob/main/demo-datasets/sample_customer_churn.csv

You can find the latest download link in the description of this video.



Best Practices for Preparing Your Data in TabH2O



Use unique column names

The first row of your dataset is treated as the header, which TabH2O uses to understand the structure and meaning of your data.



Keep your dataset clean and structured

Columns that contain no values in the selected range are automatically ignored, and fully empty rows are skipped.



Headers should always be in the first row

Make sure each column contains consistent data types and meaningful values to improve prediction quality.



Selecting a range is optional

If no specific range is selected, TabH2O automatically uses the entire sheet by default.



Well-structured data leads to better results

Organized datasets help TabH2O interpret your data more reliably and generate more accurate predictions.

Key Takeaways



The first row is your header – make it count.



Empty columns are ignored, and empty rows are skipped.



Consistent, meaningful data results in better predictions.



Let TabH2O use the full sheet or select a range as needed.

Data Structure Requirements for TabH2O Predictions



If you don't select a **specific range**, TabH2O will automatically use the **entire sheet** for analysis.



Dataset Structure Rules

- ✓ The **first row** is always treated as the header.
- ✓ Each **column** should have a **clear** and **unique name**.
- ✓ Columns containing no data in the selected range are **automatically ignored**.
- ✓ Rows that are completely empty are **skipped** as well.



Prediction Requirements

For predictions to work properly, your dataset must contain **two types of rows**:



1. Labeled Rows

These rows already contain values in the target column. They help TabH2O identify patterns and relationships within your data.



2. Unlabeled Rows

These rows contain missing values in the target column. These are the rows where TabH2O will generate predictions.

ID	Age	Income	Tenure	...	default payment next month (Target)	
24001	23	50,000	2	...	0	} Labeled Rows
24002	26	60,000	3	...	1	
24003	27	40,000	1	...	0	
...	...	...	...	...	...	
24105	29	70,000	4	...	-	} Unlabeled Rows
24106	31	55,000	2	...	-	



Important Validation Rules



You must have at least **two labeled rows**.



Your target column must contain both **filled** and **missing** values.



A fully **completed table** **will not work** for prediction generation.



Note:

A common mistake from first-time users is attempting to generate predictions on datasets where every target value is already filled in. In that scenario, TabH2O has no rows left to predict.

Example: Churn Use Case

In a churn prediction scenario, your dataset must contain both **labeled** and **unlabeled** rows for TabH2O to learn from existing data and generate predictions.

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Labeled Data (Historical Customers)

- Rows with a known churn outcome.
- Help TabH2O learn patterns and relationships in your data.



Unlabeled Data (New Customers)

- Rows without a churn outcome.
- These are the rows where TabH2O will generate predictions.

Customer_ID	Age	Tenure (Months)	Monthly Charges (\$)	Total Charges (\$)	Contract Type	Churn (Target)	
CUST_001	28	12	65.00	780.00	Month-to-month	1	Labeled (Known Outcome)
CUST_002	45	24	89.00	2,136.00	One year	0	
CUST_003	34	7	50.00	350.00	Month-to-month	1	
...	...	...	...	...	...	...	
CUST_101	30	—	60.00	—	Month-to-month	—	Unlabeled (To Be Predicted)
CUST_102	52	—	75.00	—	One year	—	
CUST_103	26	—	55.00	—	Two year	—	



Why this structure matters

Your historical customers with known outcomes act as labeled data, and new customers without outcomes are the rows you want to predict. This structure is what allows TabH2O to use your existing data as context and generate meaningful predictions for the missing values.



Summary: What You Achieved

In just a few minutes, you went from raw data to predictions—
all inside your spreadsheet.



TabH2O

 What You Did	 How It Helps	 Your Benefit
 Imported your data into TabH2O	 Data is read and understood instantly	 Start faster No setup, no exports
 Let TabH2O train the model	 Patterns are learned automatically	 Save time From weeks to minutes
 Generated predictions for missing values	 Predictions are added where you need them	 Better decisions Insights at your fingertips
 Kept everything inside your spreadsheet	 No data movement, fully private	 Secure & simple No pipelines, no complexity



Your Outcome



Minutes
instead of weeks



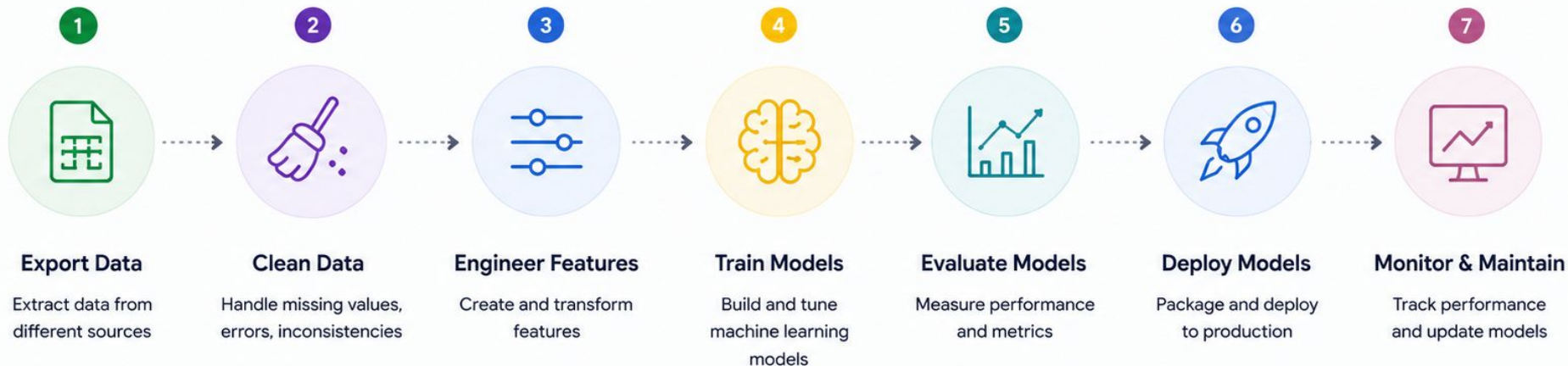
Insights
you can act on



All in one place
inside your spreadsheet

The Traditional ML Workflow

Many steps. Many tools. More complexity.



Time Consuming

From days to weeks



Requires Specialized Skills

Data, ML, and engineering expertise



Involves Multiple Tools

Disconnected workflows and environments

With TabH2O, everything is simpler.

H2O.ai



Stay in your spreadsheet

No exports. No pipelines.
No switching tools.



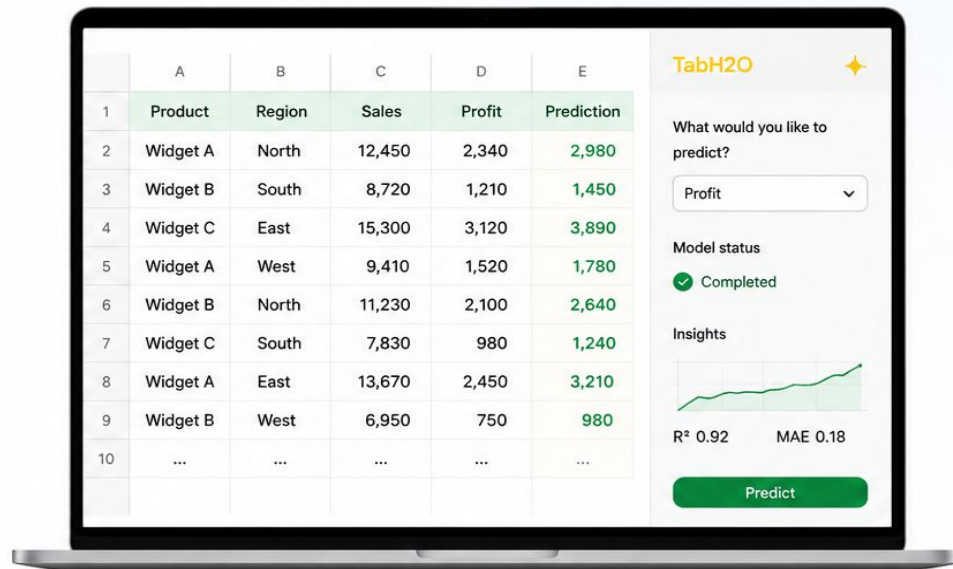
Focus on your data and your question

Ask. Predict. Understand.
All in the place you already work.



Get answers directly where you work

Predictions and insights
appear in your sheet.



Built on a foundation model approach

Delivers meaningful results even with smaller
datasets or data that is not perfectly complete.

Your data. Your control.

H2O.ai



Private by design.
Your data stays yours.



Your Data



Results

- ✓ Processed in memory
- ✓ Not stored, not cached
- ✓ Not used to retrain the model

**You stay fully in control
of your data at all times.**



Flexible and secure.
The choice is yours.

Original Data

Customer Name	City
John Smith	New York
Mary Johnson	Chicago
David Lee	Boston
...	...



Rename or
Encode (optional)

Anonymized Data

ID	Location Code
C001	NYC
C002	CHI
C003	BOS
...	...



You don't need to anonymize your data to use [TabH2O](#).



If you prefer, you can rename columns or encode values.



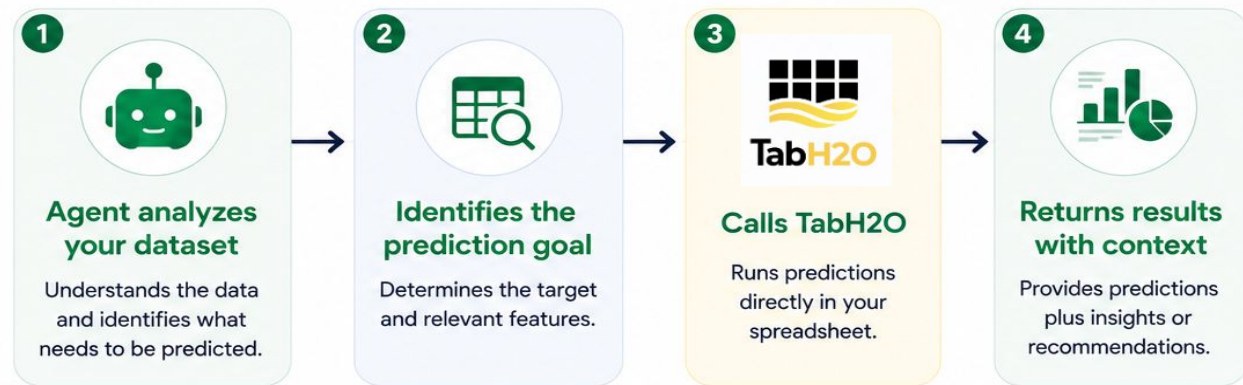
[TabH2O](#) focuses on patterns in the data, not column names – prediction quality remains the same.



Your data. Your choice. Always secure. Always in control.

Agents + TabH2O = Smarter Decisions

An agent can analyze your data, identify what needs to be predicted, call **TabH2O**, and deliver results with context or recommendations.



**More automated.
More intelligent.**

Predictions become part of a larger, data-driven decision process.



From raw data to actionable insights — all **orchestrated by your AI agent.**



Analyze



Decide



Predict



Act